The Asset Market, Money, and Prices

Lecture Notes*

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Learning objectives.

1. Define money, its functions, and how it is measured.
2. Discuss the factors that affect how people choose which assets they own.
3. Examine the macroeconomic variables that affect the demand for money.
4. Discuss the fundamentals of asset market equilibrium.
5. Discuss the relationship between money growth and inflation.

§	Topic
7-1	Money and Its Measurement
7-2	Portfolio Allocation & the Demand for Assets
7-3	The Demand for Money
7-4	Asset Market Equilibrium
7-5	Money Growth and Inflation

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Table 7.1: Sections covered.

- **The whole chapter answers one question:** what sets the price level, and why do prices rise? Everything else is machinery for that answer.
- The five goals come in order: portfolio choice → money demand → asset-market equilibrium → price level → inflation. The payoff is the last step; the first four build up to it.
- **One confusion to clear first:** to an economist "money" is *not* income and *not* wealth. It is only the assets you can spend. A billionaire with everything in land holds very little *money*.
- This is the **nominal leg** of the model. Ch 3 (labour) and Ch 4 (goods) fixed the *real* side ($\bar{Y}$ and r). Ch 7 adds P, the last piece before **IS–LM** — the goods-market (IS) and money-market (LM) model — in Ch 9.

7.1 Introduction

7.1.1 Money: a very short history

> *"Sure, I'll fix your roof — but only if you happen to have twelve dozen eggs **and** want your roof fixed."*
>
> *— barter, the hard way*

- Money is **whatever a society accepts as payment**. Its *form* keeps changing; its **three jobs** — medium of exchange, store of value, unit of account — do not.
- **Money is a *functional* category, not a physical one** (GLS = Garín, Lester & Sims, Ch. 14). We define most things by intrinsic traits — an apple is round and red — but money by what it *does*. Cash and a checking balance (a **demand deposit**) are intrinsically very different, yet both are money because both buy goods.
- **Barter** ⇒ needs a *double coincidence of wants* — slow, costly trade
  - **The joke's real point is the *double coincidence of wants*.** Barter needs both sides to want what the other has, at the same time. An economics tutor who wants a haircut must find a barber who wants an economics lesson. Money removes that search problem — which is why every society invents some form of it.
- *Did barter ever exist?* ⇒ **no pure barter economy ever found** (Humphrey) — gift/credit came first; barter = the **fallback when money breaks down**
- **Commodity money** ⇒ things with intrinsic value: cattle, salt, cowrie shells, gold (even cigarettes in POW camps)
  - Cattle gave us *pecunia* (Latin *pecus* = cattle), salt the Roman *salarium* → "salary"; then cowries, silver, gold. In WWII POW camps **cigarettes** became money (Radford, 1945) — standardized, divisible, portable, durable, accepted even by non-smokers. The catch: you can't smoke a cigarette *and* spend it, so commodity money ties up real resources.
- **Coinage** ⇒ Lydia, c. 600 BCE: the state stamps weight & purity ⇒ trade at scale
  - **First coins:** electrum in **Lydia (~600 BCE)**. **Paper money: Song-dynasty China (~11th c.)**, centuries before Europe.

- **Paper & the gold standard** ⇒ notes redeemable for metal ⇒ then **fiat money**: value by decree + trust
  - **Why does worthless paper have value?** Kiyotaki & Wright (*Journal of Political Economy*, 1989) showed fiat money is accepted only because everyone expects everyone else to accept it. Its value is a self-fulfilling convention, not intrinsic worth — and it beats barter for welfare.
  - **Gold standard** → notes redeemable for metal. The US cut the dollar–gold link in **1971**, leaving pure **fiat money** (value by decree plus trust).
- **Today** ⇒ bank balances, cards, crypto ⇒ money keeps shedding its physical form
  - **Money ≠ wealth.** Money is only the assets you can readily spend. A rich person can hold little money if wealth sits in property. **Liquidity** ranks assets by how easily they become the medium of exchange — **money is the most liquid**.
- **Three jobs, not a thing** ⇒ medium of exchange · store of value · unit of account (*numéraire*) — defined by **function, not form**
  - What each job does: **medium of exchange** ends the double coincidence, **unit of account** is the common measuring rod, **store of value** carries purchasing power forward.
  - **The three can come apart.** In hyperinflation the *store of value* fails first — people spend wages within hours — then the *unit of account* (shops quote prices in dollars). This is **currency substitution / dollarization** (Zimbabwe 2008, Argentina today).
  - **Unit of account = the *numéraire*.** Price a truck at 10,000 money-units, a beer at 1, a gun at 10, and every relative price falls out of one scale: a truck = 100 guns = 10,000 beers. Any good *could* be the numéraire; money is just the convenient one.
- **Medium of exchange** ⇒ the one job only money does ⇒ ends the *double coincidence* ⇒ **specialization → productivity**
  - **Of the three functions, medium of exchange is the one that is uniquely money's** (GLS). Houses, stocks, and bonds are *also* stores of value, and any good could serve as the unit of account; only money ends the double coincidence of wants. Pushed hard, "money" means "the medium of exchange."
  - **The big picture: money ⇒ specialization ⇒ productivity** (a link back to Ch. 3). By killing the double coincidence of wants, money enables more exchange ⇒ more specialization ⇒ the productivity gains that raise $\bar{Y}$. GLS calls a working medium of exchange "the most important thing to have developed in economic history."
- Commodity money fails each job (price swings · spoils · indivisible) ⇒ **fiat** fixes it, but lives on **belief** (over-issue ⇒ inflation tax)
  - **Why commodity money fails — one failure per function.** Independent commodity-price swings (a cattle drought, a gold strike) move *all* prices ⇒ poor **unit of account**; perishables store badly ⇒ poor **store of value**; you cannot cut a cow into 10,000 pieces to buy a beer ⇒ poor **medium of exchange**. Fiat money escapes all three (GLS Ch. 14).
  - **Fiat money's own two weaknesses ⇒ a bridge to §7-5.** It has value only while people *believe* it (the self-fulfilling convention above), and a government can always print more to pay its debts — which devalues the money and acts as an **inflation tax** on everyone holding it.

- **Was barter ever real?** Mostly a textbook myth — Caroline Humphrey (1985): *"no example of a barter economy, pure and simple, has ever been described."* Early societies ran on **gift and credit/debt**, not spot swaps (Graeber, *Debt*, 2011). Barter *does* appear, but as a **fallback**: among **strangers** with no trust, or when an existing money **collapses** (medieval coin shortages, Weimar, POW camps, 1990s Russia) — people who already knew money but briefly lacked it. (Contested — some economists argue commodity exchange still mattered.)
- *Why it matters:* once we fix what money *is*, the chapter asks what sets **the price level and the inflation rate** — money demand against the central bank's money supply.
- *Further reading:* Ferguson, *The Ascent of Money* (2008); Davies, *A History of Money*.

7.1.2 Where this chapter fits: the third market

- The book builds the economy one market at a time, then joins them in Ch 9. Three markets, three price-like unknowns: labour → $\bar{Y}$, goods → r, assets → P.
- **Ch 3 · labor market** (production function + labour-market clearing) ⇒ full-employment output $\bar{Y}$
- **Ch 4 · goods market** ⇒ desired saving = investment ⇒ real interest rate r (the IS side)
- **Ch 7 · asset market** *(this chapter)* ⇒ money demand = supply ⇒ price level P (the nominal / LM side)
- Three markets ⇒ assembled into general equilibrium **IS–LM, Ch 9**
  - **A natural question: why doesn't the asset market also set r?** It does — but only once all three markets interact together in **Ch 9 (IS–LM)**. For now r comes in from the goods market, and this chapter solves only for P. Abel defers the money-market diagram to Ch 9 for this reason.

7.1.3 What Ch 7 adds — and the payoff

- The chapter's spine is **one equation**: asset-market clearing $M/P = L(Y, i)$. Every later result — the price level, money neutrality, the inflation equation, the Fisher relation — is this one line re-solved or re-arranged. Own it and you own the chapter.
- **Portfolio choice** ⇒ how to split wealth across assets — weighing return · risk · liquidity · maturity
- **Money demand** $L(Y, i)$ ⇒ real demand ↑ with Y, ↓ with i
- **Asset-market equilibrium** $\frac{M}{P} = L(Y, i)$ ⇒ price level $P = \frac{M}{L(Y, i)}$
- **Money growth ⇒ inflation**; **Fisher** $i = r + \pi^e$ ⇒ nominal rates
- Roadmap: §7-1 Money · §7-2 Assets · §7-3 Demand · §7-4 Equilibrium · §7-5 Inflation
  - The logic climbs: **portfolio choice** → the **money-demand function** → impose **equilibrium** → read off P.
- **The payoff:** by §7-5 you can answer "if the RBI grows money 10% and output grows 3%, what is inflation?" in a single line — straight from this equation.

Bottom line. The asset market is the *nominal* leg of the model: given the real side ($\bar{Y}$ from labour, r from goods), its one job is to pin down the price level P.

7.2 Money and Its Measurement

7.2.1 What counts as money — and is there one best measure?

- Money vs non-money ⇒ a **fuzzy boundary**, not a line
  - **Money is not a category but a spectrum.** Rank assets by *liquidity*: cash → checking → savings → T-bills → house. Where you draw the "money" line is a judgement call, not a law.
- MMMFs (money-market mutual funds) ⇒ check-writing + market return ⇒ money?
  - **The MMMF puzzle makes this concrete.** A money-market mutual fund lets you write cheques (like money) yet pays a market return (like a bond). Is it money? Partly — which is why there is **no single best measure**.
- ⇒ **no single best measure** of the money stock
  - So we keep **two** rulers: **M1** (narrow, spendable) and **M2** (broad, with near-monies).
- Every asset stores value ⇒ **liquidity** (ease of spending) is the real dividing line
  - (GLS Ch. 14.) This is exactly the §7-2 portfolio axis seen from the money side: currency is the most liquid asset, a house the least, a demand deposit almost as liquid as cash.
- **Financial innovation keeps redrawing the line.** Each new near-money — sweep accounts in the 1990s, stablecoins and UPI (Unified Payments Interface) balances today — forces the aggregates to be revised. The map always lags the territory.
- **Three types of money: fiat** (value by decree — all modern currencies), **commodity** (gold, POW cigarettes — intrinsic value, but ties up resources), **cryptocurrency** (value from protocol plus belief; too volatile to serve as a unit of account).

7.2.2 The M1 Monetary Aggregate

- **M1** ⇒ currency + checkable deposits + savings accounts
  - Every part is spendable right now, so M1 is the closest measure to the textbook idea of money — note "currency" here means cash *outside* banks, plus traveller's cheques.
  - **"Currency" means cash in public hands,** not the vault cash banks hold — a common exam trip-up.
  - **A surprising fact.** US currency was ~$3,300 *per person* in 2012, yet surveys show people carry only ~$100. The rest is abroad: foreigners facing unstable currencies hold **over a quarter of M1**. The US earns an **interest-free loan** from the world, and overseas shocks show up as jumps in M1.
- Savings added to M1 in **2021** (Fed stops distinguishing savings from transaction deposits) ⇒ M1 jumps
- Series revised back only to **May 2020** ⇒ a **break**
  - **A caveat for data work.** In **April 2020** the Fed lifted Reg D's six-withdrawal cap, so savings behaved like checking and shifted **M2 → M1**. **M1 roughly doubled overnight** — a definition change, not a behaviour change. Never compare M1 across the 2020 break without a flag.
- **India angle.** Currency-to-GDP (gross domestic product) is high (~12%). The 2016 **demonetisation** of Rs 500/Rs 1000 notes pulled ~86% of currency *value* out of circulation overnight — a

recent "what is cash for?" case.

- **Liquidity ladder:** M0 (currency + bank reserves) ⊂ M1 ⊂ M2 ⊂ M3 — each step trades spendability for yield.

7.2.3 The M2 Monetary Aggregate

- **M2 = M1 +** less money-like assets
  - The added "near-monies" are very liquid but not directly spendable: savings deposits, small (< $100k) time deposits/certificates of deposit (CDs), retail MMMFs, and money-market deposit accounts (MMDAs).
- + savings deposits · small (< $100k) time deposits · retail MMMFs
  - **Why the $100,000 cutoff?** It separates household balances from institutional money. M2 tracks the spending power of ordinary households.
- Time deposits ⇒ interest + fixed term (early-withdrawal penalty)
  - That early-withdrawal **penalty** is exactly what makes time deposits less liquid than savings — liquidity is the cost of getting at your money early.
- M2 ⇒ broader, more **stable** than M1
  - **Why the stability matters:** when money moves *within* M2 (checking → savings), M2 is unchanged even though M1 moves, and that stability is what makes an aggregate useful for predicting prices.
- **A common misconception: "my credit card is money."** It isn't. A **credit card defers** payment; you settle later from your checking balance, which *is* the money. A **debit card** is just an *access device* to deposits already in M1/M2. Cards move money; they are not money.
- *(Table 7.1's Dec-2021 levels dropped from the slide — see the text for magnitudes.)*
- *Analytical Problem 1 (ABC = Abel, Bernanke & Croushore): a shift from checking to an MMMF lowers M1 but leaves M2 unchanged — the money just moved within M2.*

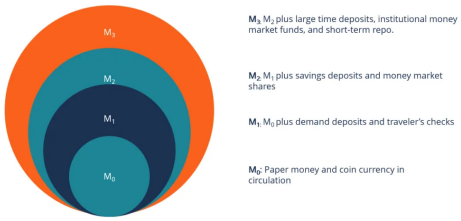
7.2.4 Elements of the money stock

Component	Role
Coins · paper money	Physical currency for everyday transactions
Checking accounts	Easily accessible funds for daily use
Savings accounts	Limited access, pays interest
Money-market accounts	Higher interest, limited withdrawals
Fixed · recurring deposits	Term savings at fixed/regular interest
Savings bonds	Government-backed savings securities

Table 7.2: Elements of the money stock.

- **The table reads top-to-bottom as a liquidity ladder** — cash at the top (instant, zero return) down to term deposits and savings bonds (higher return, least spendable). The ordering is the lesson.
- Map it to the aggregates: coins/paper/checking ≈ **M1**; savings, money-market, fixed/recurring deposits ≈ **M2 and broader**.

- **For Indian students:** "fixed deposits (FDs)" and "recurring deposits (RDs)" are the local names for time deposits; "savings bonds" are like RBI Floating Rate Bonds and small-savings schemes. The *category* — term savings, interest-bearing, low liquidity — is the same even where names differ.
- The chain again: **M0** ⊂ **M1** ⊂ **M2** ⊂ **M3** — each step adds yield and loses liquidity.



The aggregates nest. Each layer adds less-liquid, higher-yield assets: M_0 (cash) ⊂ M_1 (+ demand deposits & traveller's cheques) ⊂ M_2 (+ savings & money-market shares) ⊂ M_3 (+ large time deposits, institutional funds, repo). Where you draw the "money" line is a judgement call.

7.2.5 How does a central bank change the money supply?

- **What "creating money" really means.** When the central bank buys a bond it does not print notes — it **credits the seller's bank reserve account with a keystroke**. Money is created by a balance-sheet entry.
- **Money supply = money stock** available in the economy
- **Buy** assets from the public ⇒ supply ↑ (open-market *purchase*)
- **Sell** assets to the public ⇒ supply ↓ (open-market *sale*)
- Together ⇒ **open-market operations**
  - The mechanism is **reserves**: a purchase puts new reserves into the system, a sale drains them out — open-market operations are a central bank's main day-to-day tool.
  - **How it works today:** modern central banks target a *price* (an interest rate), not a *quantity* of money, and let the quantity adjust. OMOs are how they hit the rate.
- **India:** the Reserve Bank of India's (**RBI**) Monetary Policy Committee (**MPC**) sets the **repo rate**; the Liquidity Adjustment Facility (**LAF**, repo/reverse-repo) plus **OMOs**, Cash Reserve Ratio (**CRR**) and Statutory Liquidity Ratio (**SLR**) move the money stock. CRR and SLR are quantity tools India still uses actively, unlike the US.
- **US:** the **Fed** (1913) acts through the Federal Open Market Committee (**FOMC**); its **dual mandate** is stable prices plus maximum employment, and it is also **lender of last resort**. Other tools: policy rate, reserve requirements, **forward guidance**, quantitative easing/tightening (**QE/QT**), discount window.

Policy Application: financial innovation and the money aggregates

Technology keeps redrawing what counts as money and how much people hold. ATMs and debit cards, bank *sweep* programs that move checking balances into savings overnight (which shrinks measured M1 and once complicated the Fed's reserve targeting), and today instant mobile rails such as India's UPI all shift money demand and blur the aggregates. Each wave makes a given money-growth number harder to read — one more reason central banks target an interest rate rather than the quantity of money.

7.2.6 Central Banks and Government Financing

- The central bank (CB) buys **new govt bonds directly** from Treasury ⇒ financing spending by **printing money**
  - **The key distinction.** Buying bonds in the *secondary* market (from the public) is ordinary monetary policy. Buying **new bonds *directly* from the Treasury** is **monetising the deficit** — financing the government by printing money.
- Common abroad ⇒ **forbidden by law in the US**
  - **Why it is banned or fenced off.** Direct monetisation erases the government's budget constraint and invites inflation, so walling it off protects **central-bank independence** and price stability. **India:** the RBI funded deficits through *automatic* ad-hoc Treasury bills until this was **phased out by 1997** and replaced with capped Ways-and-Means Advances.
- Notation: M = money supply (M1 or M2)
- **Seigniorage = revenue from creating money,** and it is an **inflation tax**: new money erodes the real value of cash people already hold. It is **regressive** — the poor hold a larger share of their wealth as cash, so they bear more of the tax (Pérez, US Survey of Consumer Finances data).
- **The warning (and bridge to §7-5):** when a weak tax system — war, state collapse — pushes a government onto seigniorage, you get the hyperinflations covered later. The disease and the cure are the same lever.

Bottom line. "Money" is a spendability spectrum, not one thing: M1 (narrow) and M2 (broad) are two rulers, and the central bank moves the stock through open-market operations.

7.3 Portfolio Allocation & the Demand for Assets

7.3.1 How do people choose which assets to hold?

How do people allocate wealth among assets?

- *Background* — **the micro-foundation of money demand:** money is one asset competing for space in a portfolio, so why anyone holds low-return money needs the rules of portfolio choice first.
- **Expected return**
- **Risk**
- **Liquidity**
- **Time to maturity**
- **The four trade off — no asset wins on all of them.** Money wins on liquidity and loses on return; stocks are the reverse.
- **The one "free" move is diversification** — spreading wealth across weakly-correlated assets cuts risk *without* cutting expected return (Markowitz; the only "free lunch" in finance).
- A good self-test: place cash, an equity-index SIP (systematic investment plan), a bank FD, and a house on all four factors — doing so rebuilds the whole section.

7.3.2 Expected Return

- **Rate of return** = asset's value gain per period
  - Bank account ⇒ interest rate; stock ⇒ dividend + price gain
  - **Worked example:** buy at Rs 100, sell at Rs 108, Rs 2 dividend ⇒ a 10% return — a stock's return is **dividend yield + capital gain**, a bank account's is just the interest.
- What matters ⇒ **expected real** return
  - *Expected*, because the future is uncertain; *real*, because inflation eats nominal gains — a 12% nominal FD in 7% inflation is only a **5% real** return.
- Investors prefer **high** expected return, other things equal
- Money has the lowest return — held mostly for its **liquidity** services
  - **A puzzle for §7-3:** if investors like high returns, why hold money, which returns ~0? Because they are buying **liquidity**, not return — and the price of that convenience is the forgone interest.

7.3.3 Risk

- **Risky** ⇒ significant uncertainty about the return
  - Not "losing money" — *uncertainty*. A government bond and a lottery ticket can share the same *expected* return; the lottery is riskier because its outcomes are more spread out.
- Investors dislike risk ⇒ hold only if **expected return higher**
- **Risk premium** = extra expected return on a risky asset over a comparable **safe** asset.
  - Read it as a price: what the asset must pay to persuade a risk-averse investor to hold it — so the riskier the asset, the larger the premium.
  - **The number (US, 1871–2021):** stocks ~**8.5%/yr** real vs bonds ~**2.7%**, a ~6-point **equity premium** — so large that ordinary risk aversion cannot explain it (the **equity-premium puzzle**, Mehra–Prescott, 1985).
  - **Who built the theory:** Markowitz, Miller & Sharpe shared the **1990 Nobel** for the risk–return trade-off and the Capital Asset Pricing Model (CAPM). A behavioural footnote: people feel losses about twice as much as equal gains (*loss aversion*).

7.3.4 Liquidity

- **Liquidity** = ease/speed of exchange for goods, services, or assets
  - Add a second test to the definition: not just speed, but whether you can sell *without moving the price against you*. Cash passes both; a house passes neither in a hurry.
- Money ⇒ **very** liquid; stocks/bonds ⇒ fairly liquid
  - **Ranked:** cash (perfect) → bank deposit → listed stock/bond (seconds, tiny spread) → house (weeks to months, 1–2% fee, a haircut if you rush) → family business (maybe no buyer at all).
- House ⇒ **illiquid** (slow, costly to sell)
  - **Illiquidity you can feel:** needing cash fast with wealth "all in the house" can force a fire sale costing 10–20% — the fear that makes people hold some low-return money as a buffer

(the precautionary motive).
  - Because liquidity is valuable, illiquid assets pay a **liquidity premium** — the flip side of why money pays so little.
- All else equal ⇒ prefer liquid assets

7.3.5 Time to Maturity

- **Maturity** = time until all promised payments are made
  - The anchor fact: money has **zero** maturity; a 30-year bond sits at the far end of the same axis.
- Long-term bonds ⇒ generally **higher return**
  - **Expectations theory:** in equilibrium, bonds of different maturity give the *same expected return* over a holding period (else arbitrage), so a long yield ≈ the *average* of expected future short rates.
- Long-term prices swing **more ⇒ riskier**
  - **Why the extra return isn't free:** long rates sit *above* short because long bonds' prices swing more when rates move — that compensation is the **term (risk) premium**.
- *Numerical Problem 1 (ABC): expectations theory + term premium → the bond yield curve.*
  - **The yield curve predicts recessions.** When it **inverts** (short > long) it has preceded almost every US recession; markets invert it when they expect rate *cuts*, i.e. a slowdown.
  - *Worked (NP1):* one-year 6%, two-year 5.55%, three-year 5.33% ⇒ an inverted curve.

7.3.6 Types of Assets and Their Characteristics

- **Money (M1)** ⇒ most liquid, low return
- **Bonds** ⇒ less liquid, higher return, riskier (many maturities)
- **Stocks** ⇒ fairly liquid, can be very risky
- **Houses** ⇒ illiquid + risky
  - **Case — the 2007 housing crisis.** In the 2000s rising prices made housing look safe and high-return. Lenders wrote **subprime** mortgages with low teaser rates that later **reset** higher. The logic held only while prices rose. When they stalled, defaults spread through **mortgage-backed securities** into the 2008 crisis. Foote–Gerardi–Willen (2012) read it as optimistic bets that turned out wrong — a bubble, not a conspiracy.
- The ladder on all four factors: **money** (liquid, ~0 return, safe) → **bonds** (more return, more risk, less liquid) → **stocks** (dividends + capital gains, much riskier) → **housing / small business** (illiquid *and* risky, possibly high return).
- **Life-cycle angle:** the young can bear **risk** (decades to recover); retirees need **income and liquidity**. Same theory, opposite portfolios.
- **A preview of §7-4:** the **sum of all asset demands = total wealth**. This budget constraint delivers **Walras' Law** later — *across all markets the excess demands sum to zero*, so if money demand is met, bond demand must be too.

Data Application: the 2007–08 housing crisis

Was the crisis fraud, or honest error? Foote, Gerardi & Willen (FRBB, 2012) read it as *ex post* bad decisions by lenders, borrowers, and investors alike, all betting in an apparent bubble that house prices would keep rising. When prices stalled, subprime defaults spread through mortgage-backed securities into the 2008 collapse. The portfolio-choice lesson: housing is both risky and illiquid, and "safe and high-return" is usually a warning sign.

Bottom line. Every asset is judged on return, risk, liquidity, and maturity; money wins only on liquidity, which is why people hold it despite its near-zero return.

7.4 The Demand for Money

7.4.1 The Demand for Money

- **Money demand is a *derived* demand** — nobody wants money for itself, only for what it does (easy transactions). The real question: how much *liquidity* is worth buying, given that it costs forgone interest?
- **Money demand** = monetary assets people want to hold
- Depends on **expected return · risk · liquidity**
  - The same three factors apply, but money's profile is fixed: **lowest return, lowest risk, highest liquidity.** The decision reduces to one trade-off — *liquidity now vs interest forgone.*
- Money ⇒ most liquid, low return ⇒ held for **liquidity**
  - **The key price: the nominal rate i is the opportunity cost of holding money** — what you give up by sitting in cash instead of bonds. This drives the whole money-demand curve.
- Key macro drivers ⇒ **price level · real income · interest rates**
- **Cambridge equation ($M^d = kPY$):** people hold a fraction k of nominal income as money, with $k = 1/V$. Same content as the quantity theory, seen from the *demand* side — "how many weeks of spending do I keep on hand?"

7.4.2 A higher price level raises nominal money demand one-for-one

- **The mechanism is purely mechanical.** The *same basket* of real transactions costs more rupees when prices are higher, so you need proportionally more cash.
- Higher price level ⇒ **more money** to transact
- Prices **double** ⇒ nominal money demand **doubles** ⇒ *proportional*
  - **Historical anchor:** US prices are ~10× their 1935 level, so the same real transaction needs ~10× the money. Double *all* prices ⇒ nominal money demand doubles exactly — price elasticity = **1**.
  - **Worked example:** a family spending **$520/month** holds ~**$260** on average (cash runs down between paydays). Double all prices ⇒ spends $1,040 ⇒ holds ~$520. Real holdings are unchanged.
- **Real** money demand M^d/P ⇒ **unchanged** by the price level

- **The key assumption — no money illusion.** People care about *real* balances M^d/P, not the rupee count. Caveat: in the short run people sometimes *do* feel richer when a nominal wage rises even if prices rose as much, but the long-run theory here rules that out.
- **Signpost — keep the three "moneys" straight:** *nominal money* $M^d = P\,L$ (the rupee count), *real money demand* $L(Y, i) = M^d/P$ (its purchasing power), and *monetary assets* (the spendable bucket of wealth, §7-4). From here, "money demand" means the **real** one unless we say nominal.

7.4.3 Money demand rises with income — but less than proportionally

- Real income ↑ ⇒ more transactions ⇒ money demand ↑
  - The rise is **less than one-for-one** — a subtle, exam-favourite result.
- But the rise is **less than proportional** to income
  - **Why less than proportional? Economies of scale in cash management.** By the Baumol–Tobin "inventory" model, managing money has fixed costs, so richer and more financially sophisticated people economise: they park idle cash in interest-bearing assets and convert in fewer, larger moves. Double your income and average cash less than doubles.
  - **So income elasticity is positive but < 1** — about **0.5–0.7** for M1, closer to **1** for M2. Goldfeld's classic estimate is **2/3** (used in §7-5).
- **Why it matters:** with elasticity < 1, money demand grows slower than income as an economy grows, so **velocity** — how often each rupee is spent in a year — **trends up** with development. That is the hinge for forecasting inflation.

7.4.4 How do interest rates move money demand?

- i on **nonmonetary** assets ↑ ⇒ money demand ↓
  - Two *opposing* rates pull on money demand: a higher nonmonetary i raises the forgone interest on cash (demand ↓), while a higher i^m paid *on* money makes it more competitive (demand ↑).
- i^m on **money** ↑ ⇒ money demand ↑
- What matters ⇒ the spread $i - i^m$
  - Read the spread as the net penalty for staying liquid: if banks paid the bond rate on checking, that penalty would vanish and money demand would explode.
  - **Why we usually just write i:** money pays little ($i^m \approx 0$) and barely moves, so the spread $\approx i$. We also treat the many nonmonetary rates (T-bills, corporate bonds, mortgages) as one i because they move together.
- **Real-world link:** this is why rising policy rates pull money out of current accounts into FDs and liquid funds — households respond to a wider spread.

7.4.5 The Money Demand Function

$$M^d \quad = \quad P \times L(Y, i) \qquad (7.1)$$

- **(7.1) in words:** nominal money demand = the price level **times** the real liquidity people want, $L(Y, i)$. P sits *outside* L because of the one-for-one price result; Y and i sit *inside* because their effects are not proportional.
- ↑ in price level P and real income Y; ↓ in i
  - The reasons behind the signs: L rises with Y because more transactions need cash, falls with i because of the higher opportunity cost — and nothing else enters the core function.
- Real form ⇒ $M^d/P = L(Y, i)$ (7.2)
  - This is the form we actually use: it drops the price level and speaks of *real balances* (purchasing power held as money).
- With the spread ⇒ $M^d/P = L(Y,\ i - i^m)$ (7.3)
  - **(7.3)** just makes the spread $i - i^m$ explicit; i^m is dropped from (7.1) because it varies little.
- **Empirical note:** a **log-linear** form, $\ln(M/P) = a + \eta_Y \ln Y - \eta_i\, i$, fits well and reads coefficients straight off as elasticities — the source of the next slide's values.

7.4.6 Other Factors Affecting Money Demand

- **Wealth** ↑ ⇒ money demand ↑ (but wealth moves slowly ⇒ demand stable)
  - The effect is only *slight*, so quarter-to-quarter money demand is fairly **stable** — the exception being bubbles and crashes, which move wealth fast.
- **Risk** ↑ ⇒ flight to money (financial crisis, 9/11)
  - **Risk cuts both ways.** Risk in *other* assets ↑ ⇒ **flight to money** (the "dash for cash" of 2008, 9/11, March 2020). But risk *in money itself* — erratic inflation — ⇒ demand **falls** as people flee to real assets or foreign currency. Which effect wins tells you whether the panic is *financial* or *monetary*.
- **Liquid alternatives** ↑ ⇒ money demand ↓ (esp. M1)
  - This bites M1 hardest: deregulation gave bonds and funds cheque-like features, so they poached transaction balances.
- **Payment tech** (cards) ⇒ less cash needed (~70% of Americans use cards; 1.2 bn in circulation)
  - This is the big modern *destabiliser* of money demand, and **India's UPI is the clearest example** — instant digital payments let people hold far less idle cash, shifting demand faster than any past deregulation. A good question: has UPI made the RBI's money-demand forecasts harder? (Yes.)
- *Recessions, crises & money demand:* real money demand is **procyclical** (Y and i both are). In a crisis a **risk-driven surge** can push it *above* what income and rates predict — a flight to safety that **amplifies** the downturn.
- *Financial crises & policy:* **2008 Lehman** → flight to quality (wealth ↓, other assets look risky, rates → 0, income ↓) ⇒ a jump in money demand that the Fed **accommodated** via QE. This echoes **Friedman–Schwartz**: failing to meet money demand turned the 1930s slump into the

Great Depression.

7.4.7 Elasticities of Money Demand

- **Elasticity** = % change in money demand per 1% change in a factor
  - Because it is *unit-free*, income, interest, and price effects all sit on one comparable scale.
- **Income** $\eta_Y \Rightarrow$ **positive but** < 1 (Goldfeld $\approx 2/3$)
  - Rule of thumb (Goldfeld, $\eta_Y = 2/3$): **3% real growth needs ~2% more real money** to keep prices steady — the number used in the inflation forecast (§7-5).
- **Interest** $\eta_i \Rightarrow$ small, **negative**
  - Only **a few tenths**, so rate hikes trim money demand just a little and the curve is fairly steep in i.
- **Price** $\eta_P \Rightarrow$ **unitary** $(= 1)$
  - Money demand scales one-for-one with P — the result baked in earlier.
- **Why they matter:** these three numbers are the **policy dials** — they let a central bank set a **non-inflationary money-growth rate** (§7-5) and gauge how a rate change moves demand.
- **A caveat:** not laws. After ~1974 the standard equation began **over-predicting** money holdings — Goldfeld's "**missing money**" puzzle — so the Fed eventually stopped *targeting* money.

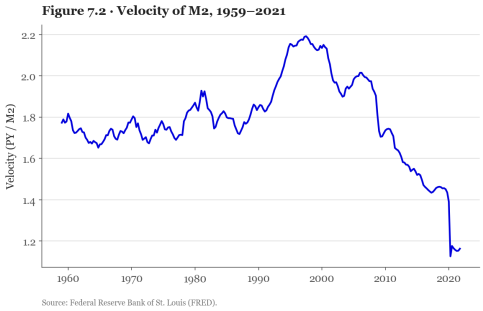
7.4.8 Velocity of Money

$$V = \frac{\text{nominal GDP}}{\text{nominal money stock}} = \frac{PY}{M} \qquad (7.4)$$

- **Velocity** $\Rightarrow$ how often money "turns over" each period
  - Put in numbers: if M2 = Rs X and nominal GDP = Rs 5X, each rupee of M2 is spent ~5 times a year on final goods.
  - **Intuition for** $M/P = Y/V$ **(Pérez):** real balances are the *goods your money can buy*. If the only good is brisket, the economy has \$100, and brisket is \$20/lb, then real balances = **5 lb of brisket**. Money is a claim on stuff.
  - **What high** V **signals:** money racing around because people *don't want to hold* it — V rises when holding cash is costly (high i, high inflation; in hyperinflation it **explodes**) or easy to avoid (UPI, cards), and falls when people want the safety of cash (recessions, near-zero rates — V fell sharply in 2020).
- M2 velocity $\Rightarrow$ long **downward trend**
- Unstable velocity $\Rightarrow$ money growth a **poor predictor** of nominal GDP
  - **Why this section exists:** the money→prices link is reliable only *if* V *is stable*. **M2 velocity trended down** (financial innovation, Fig. 7.2); **M1 velocity has been erratic since ~1982**. Unstable V makes a given money-growth number map to an unpredictable nominal GDP — which is why central banks **dropped money targeting** for interest-rate rules.
- *Numerical Problem 2 (ABC): compute* V *from a money-demand function.*

- **Worked value (NP2):** with $M^d/P = 500 + 0.2Y - 1000i$, $Y = 1000$, $i = 0.10 \Rightarrow$ real demand = 600, so $V = Y/(M/P) = 1000/600 = \mathbf{5/3}$. V falls out of the demand function, not from thin air.

- **Two forms:** transactions $MV = PT$ (T = volume of transactions); income $MV = PY$. In growth rates, **%ΔM + %ΔV = %ΔP + %ΔY** — the bridge to the inflation equation.



Velocity is not constant. M2 velocity rose into the late 1990s, fell steadily afterwards, and dropped sharply in 2020. The money→prices link is only as stable as V — which is exactly why the Fed stopped targeting the monetary aggregates.

Take-away: the data *refute* the quantity theory's constant-V assumption, so "inflation = money growth" is a long-run statement, not a short-run one.

Data Application: the P* model and the M2 puzzle

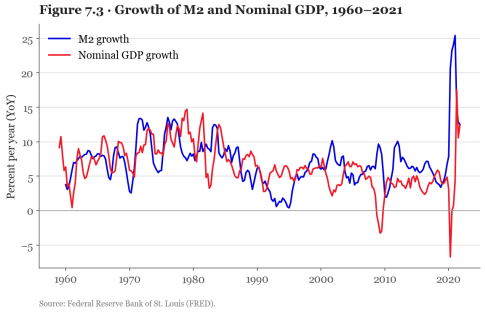
When a stable velocity almost became a policy rule. Because M2 velocity once looked stable, Federal Reserve economists built a model in which the long-run price level, P^*, is tied mainly to the level of M2 (Hallman, Porter & Small, *AER*, 1991). The irony is sharp: almost as soon as the paper appeared, M2 velocity began to wander and the Fed quietly deemphasised M2 for policy.

7.4.9 The Quantity Theory of Money

$$\frac{M^d}{P} = kY \qquad (7.5)$$

- **Quantity theory** ⇒ real money demand ∝ **real income**
  - The move that defines it: fixing velocity ($k = 1/V$ constant) *assumes away* the interest effect entirely.
- Assumes **constant velocity** (no income/interest effect)
  - With V fixed, M and nominal GDP move together and money growth maps straight to inflation. Everything rides on this single assumption, so it must be tested against data.
- **But** M1 velocity ≠ constant ⇒ rose 1960–80, erratic since
  - **Where it works:** strong in the **long run** and in **high inflation** (money growth swamps all else); weak in the **short run**, where V wanders. Hence the Fed dropped M1 targets (1987) and de-emphasised M2 (1993).
- **M2 velocity** ⇒ closer to constant, **not** short-run
- **Origin:** one of the oldest ideas in economics. **Jean Bodin (1560s)** blamed Europe's "price revolution" on New World silver — the first clear money→prices argument. Lineage: **Copernicus → Bodin → Hume → J.S. Mill → Fisher → Friedman.** Hume's version already has **money neutrality**: double the money, double all prices, change nothing real.
- **Growth form:** $\pi \approx g_M + g_V - g_Y$; with constant V, **price stability needs** $g_M = g_Y$ — grow money as fast as output, no faster.
- *Bitcoin & crypto:* a decentralised digital asset on a **blockchain**, no issuer; **fixed supply** (21 m BTC) caps inflation but drives wild **volatility**, which, plus high fees, makes it a poor unit of

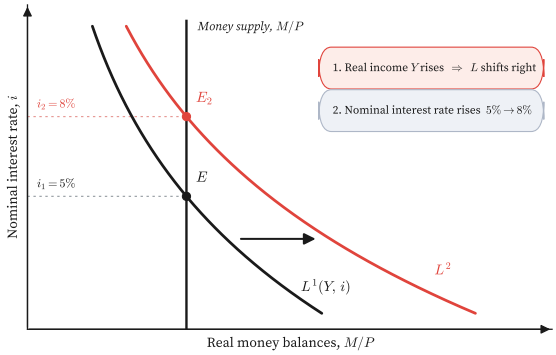
account or medium of exchange. Fixed supply + variable demand = unstable price — the quantity theory in reverse.



Money growth and nominal GDP move together. Their gap is the growth of velocity: when M2 growth runs persistently above nominal-GDP growth, velocity is falling. Short-run divergences are large; the long-run co-movement is the quantity theory.

Take-away: the long-run co-movement *confirms* $MV = PY$; the short-run gaps are precisely the velocity shifts the quantity theory assumes away.

7.4.10 What sets the interest rate? — a preview (full story in Ch. 9)



- **What the diagram shows:** real money supply M/P (vertical — fixed by the CB) against real money demand $L(Y, i)$ (downward-sloping in i). Their crossing pins the nominal rate i.
- **Why demand slopes down:** higher i = higher opportunity cost of holding money ⇒ people economise on cash ⇒ quantity falls. Same as any demand curve, but the "price" on the axis is the forgone interest.
- **Why supply is vertical:** at a point in time the CB and the price level fix real balances, so quantity does not respond to i.
- **The build:** $L^1 \to M/P \to$ equilibrium E ($i_1 = 5\%$) $\to$ raise income so demand shifts right ($L^1 \to L^2$) $\to$ with supply fixed, the market clears at a higher rate ($i_2 = 8\%$). "More income ⇒ higher interest rate" is the seed of the **LM curve**.
- **A caveat:** Abel defers this diagram to Ch 9, because putting i on the axis quietly assumes the goods market has already set r. You cannot solve i and r one at a time — they come together. Treat this as intuition only; the rigorous version is general equilibrium.

Bottom line. Real money demand $L(Y, i)$ rises with income (less than one-for-one) and falls with the nominal rate; velocity is its flip side and is *not* constant.

7.5 Asset Market Equilibrium

7.5.1 Asset Market Equilibrium — An Aggregation Assumption

- Group all assets into **two** categories:
  - **The trick that makes the model solvable:** collapse thousands of assets into just **two** — *money* and *everything else*. Two assets, one budget constraint, so clearing one market clears the other (next slide).
- **Money** ⇒ currency + checking; pays i^m; supply fixed at M
  - Note $i^m \approx 0$ in practice, and the supply M is fixed by the central bank, not the market.
- **Nonmonetary** ⇒ stocks, bonds, land…; pays $i = r + \pi^e$ (Fisher relation — §7-5); supply fixed at NM
- **The "fixed supply" caveat:** supplies are fixed only in the **short run** — fine for a snapshot where M is set and the stock of bonds and capital has not changed. Over longer horizons both move (later chapters).
- **Why this crude step is safe:** only *one margin* matters — money vs non-money — because that is what pins the price level. Detail inside "non-money" is irrelevant for P.

7.5.2 If the money market clears, the whole asset market clears

- Demanded: $M^d + NM^d$ (7.6) = Supplied: $M + NM$ (7.7)
- Subtract ⇒ $(M^d - M) + (NM^d - NM) = 0$ (7.8)
  - With only two assets this is decisive: if the **money market** clears ($M^d = M$), the **bond market clears automatically** ($NM^d = NM$). Two markets for the price of one.
- ⇒ excess demands sum to **0** (Walras' Law)
  - **Walras' Law:** since everyone's asset demands add up to their wealth, and supplies add up to that same wealth, the **excess demands across all markets sum to zero**. The last market has no freedom — if every other market clears, it must clear too.
- Money market clears ⇒ **entire** asset market clears
  - **Why it matters:** we can study the whole asset market through one condition, $M/P = L(Y, i)$ — exactly the **LM curve** of IS–LM. That is the payoff of the two-asset step.
- **Concretely:** if households want Rs 10 more money than exists, they must want Rs 10 *fewer* bonds than exist — the bond market holds the mirror-image excess. Fix one and you fix both.

7.5.3 The price level = nominal money supply ÷ real money demand

$$\frac{M}{P} = L(Y,\ r + \pi^e) \quad (7.9) \qquad \Longrightarrow \qquad P = \frac{M}{L(Y,\ r + \pi^e)} \quad (7.10)$$

- (7.9) ⇒ **real supply = real demand**
  - This is the chapter's **hinge equation** — every later result is just $M/P = L(Y, r + \pi^e)$ re-solved or re-arranged.
- M ⇒ central bank; π^e fixed (for now)

- $Y \Rightarrow$ labor market + production; then $r \Rightarrow$ goods market
  - **The order of determination:** Y comes from the **labour market + production** → given Y, r comes from the **goods market** → M is set by the **central bank** and π^e is taken as given → the only unknown left in (7.9) is P. The asset market's job is to solve for the price level, nothing else.
- (7.10) $\Rightarrow$ P = nominal supply ÷ real demand $\Rightarrow$ double $M \Rightarrow$ double P
  - The key rider: doubling M doubles P *with no real variable touched* — the first glimpse of money neutrality.
  - **Worked number:** if $M/P = L = 10$ and $M = 300$, then $P = 30$; raise M to 600 and $P = 60$. The ratio does all the work.
- *Numerical Problems 3 & 4 (ABC): solve for the equilibrium interest rate and the price level.*
- **This is the classical dichotomy / money neutrality** — the most important long-run idea in the chapter. Nominal things (M, P) and real things (Y, r, real wage) live in separate rooms. It is why printing money cannot make a country richer, only more inflationary.

Theoretical Application: money neutrality and the classical dichotomy

In the classical model the real variables — output, the real wage, the real interest rate — are pinned down by technology, preferences, and resources, while the money supply fixes only the *level* of nominal variables. Double the money and, in the long run, all prices and nominal values double while every real quantity is unchanged. That is **money neutrality**: it is why the asset market here solves for P alone, and why printing money cannot raise a country's real income.

Bottom line. One condition, $M/P = L(Y, i)$, fixes the price level: $P = M/L$. Double the money and you double the price level — money is neutral.

7.6 Money Growth and Inflation

7.6.1 Does money growth cause inflation?

$$\frac{\Delta P}{P} = \frac{\Delta M}{M} - \frac{\Delta L(Y, r + \pi^e)}{L(Y, r + \pi^e)} \qquad (7.11)$$

- Inflation closely tied to the **growth rate of the money supply**
  - **The headline:** if real money demand is **stable**, **inflation ≈ money growth**. This is the core monetarist claim — Friedman's "inflation is always and everywhere a monetary phenomenon."
  - **The evidence (Pérez):** the money-growth/inflation correlation is ~**0.77** across US decades and ~**0.78** across countries — tight, and tighter the higher the inflation.
- Equilibrium $\Rightarrow$ **inflation = money-supply growth − real-money-demand growth**
  - Nothing new here: **(7.11) is just (7.10) in growth rates** — take percentage changes of $P = M/L$ and you get it.
- **A subtlety: in hyperinflation money growth can *outrun* inflation.** Zimbabwe 2005–07: consumer price index (CPI) **5,316%** but M2 growth **9,914%**. Why faster than prices? Real money *demand collapses* — people dump cash as fast as they earn it, so V explodes and the real-demand

term goes sharply **negative**. The flight then accelerates inflation further. It is a self-feeding loop, not a clean one-for-one.

- *Transition Europe:* after 1990, Russia and others ran high inflation from **deficit monetisation** while disciplined countries stayed low. Across countries the relationship is roughly a **45-degree** line.

Data Application: what causes inflation?

Over short horizons inflation has many drivers — supply shocks, demand swings, exchange-rate moves. Over the long run the cross-country evidence is clear and agrees with Larry Ball's survey (Federal Reserve Bank of Philadelphia, 1993): sustained inflation tracks money growth, and the fit is tightest where inflation is highest.

7.6.2 How do we forecast inflation from money growth?

$$\pi = \frac{\Delta M}{M} - \eta_Y \frac{\Delta Y}{Y} \qquad (7.12)$$

- Long run ⇒ i constant ⇒ watch only growth in Y (η_Y = income elasticity)
  - **Long-run simplification:** with i roughly constant, real money demand grows only with **income**, so (7.11) collapses to **(7.12)** $\pi = \Delta M/M - \eta_Y \, \Delta Y/Y$ — inflation = money growth minus income-driven demand growth.
- **Your turn:** with $\Delta M/M = 10\%$, $\Delta Y/Y = 3\%$, $\eta_Y = 2/3$ — what is π?
  - **The "Your turn" answer:** $\Delta Y/Y = 3\%$, $\eta_Y = 2/3$ ⇒ real demand grows $2/3 \times 3\% = 2\%$; with $\Delta M/M = 10\%$ ⇒ $\pi = 10\% - 2\% =$ **8**%. Try it before reading the answer.
- *Numerical Problems 5–7 (ABC): use elasticities to predict inflation and the price level.*
- **India check:** if the RBI grows M3 ~10%/yr and real GDP grows ~6–7%, (7.12) predicts mid-single-digit inflation — close to the RBI's 4% (±2) target band. The framework is not just a US toy.
- **A caveat:** this is a *long-run* forecast. In the short run i and V move, so it anchors the *trend*, not next quarter's CPI.

Application: forecasting trend inflation

A one-line policy check (India). Long-run inflation is about money growth minus income-driven money demand, $\pi \approx g_M - \eta_Y \, g_Y$. With broad money growing ~10% a year, real output ~6–7%, and income elasticity near 2/3, the formula gives mid-single-digit trend inflation — inside the RBI's 4% (±2) band. The same arithmetic flags danger abroad: once a government leans on seigniorage and g_M runs into the hundreds of percent, the output and velocity terms become rounding error and inflation tracks money growth almost exactly.

7.6.3 In high inflation, money-supply growth dominates

- Both demand & supply growth matter, **but supply dominates** in high inflation
  - **The payoff:** this is *why* the quantity theory, loose in the quiet short run, is nearly exact in high inflation — the money term dominates so much that demand swings are rounding error.
- Real demand moves little ($\eta_Y = 2/3$: ±15% growth ⇒ only ±10% demand)

- **The point is about *variance*, not levels.** Real money demand barely moves: with $\eta_Y = 2/3$, even a ±15% swing in income changes real demand by only ±10%. The demand term stays in a narrow range.

- Supply growth differs by **hundreds of points** ⇒ explains inflation gaps
  - **Money-supply growth ranges over *hundreds* of percentage points** across countries. When one term varies by ±10 and the other by ±900, the big one wins — so inflation differences are **almost entirely a money-supply story**.
- **Policy lesson:** hyperinflation cannot be blamed on "greedy shops" or "supply shocks." The arithmetic forces it back to the printing press.

7.6.4 Why Allow Rapid Money Growth?

- Why permit it if it causes inflation?
- ⇒ sometimes the **only way to finance** government spending
  - **The question is really fiscal.** Rapid money growth is rarely a monetary *mistake*; it is usually a **fiscal choice**. When a government cannot tax enough and cannot borrow (war, collapse, no credibility), the printing press is the **last-resort financier**. The inflation is the symptom; the broken budget is the disease.
- Esp. **very poor** or **political-crisis** countries (seigniorage)
  - **The numbers (Fischer, 1960–78):** seigniorage was ~**3%** of revenue in the US but ~**46%** in Argentina, whose inflation ran ~57%. Reliance on the press and inflation rise together.
- **A self-defeating twist:** there is a **Laffer curve for seigniorage**. Push money growth too hard and people flee cash so fast that *real* revenue from printing *falls*. Hyperinflating governments print more and more to collect less and less (Cagan).
- **Political economy:** benefits are concentrated and immediate, costs diffuse and delayed — which is why money is delegated to **independent central banks** insulated from the treasury.
- *The social costs of inflation (Mankiw §6-5, Pérez):*
  - **Expected inflation: shoeleather** costs (minimising cash), **menu** costs (reprinting prices), **relative-price distortions**, and **tax distortions** (nominal capital gains taxed ⇒ real over-taxation).
  - **Unexpected inflation** adds **arbitrary redistribution** (debtors gain, creditors lose when $\pi > \pi^e$) and **greater uncertainty**, which itself chokes off long-term contracts and investment.

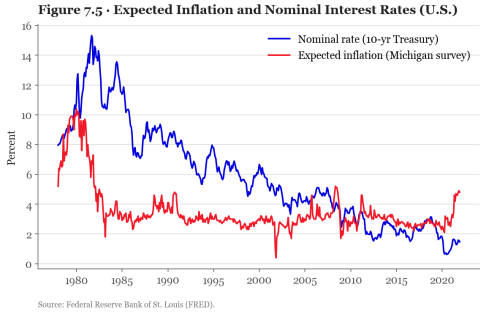
7.6.5 Expected inflation raises the nominal interest rate

$$i = r + \pi^e \qquad \text{(Fisher relation)}$$

- Expectations **can't** be ignored in the real world
- $\pi^e > 0$ ⇒ **raises the nominal rate** ⇒ changes money demand & P
  - **Fisher $i = r + \pi^e$:** lenders demand compensation for the *expected* erosion of their money, so the nominal rate carries an inflation premium. Higher π^e ⇒ higher i ⇒ (via money demand) feeds back into P. This is why we finally *endogenise* π^e instead of fixing it.

- Stable conditions ⇒ $\pi^e \approx \pi$ (actual)
  - **A puzzle (Pérez):** if i responds to *expected* inflation, why does it track *actual* inflation so well? Because **inflation is highly persistent** ($\pi_t = \alpha + \rho\pi_{t-1} + e_t$, $\rho \approx 0.77$ before 1985, ~0.51 after), so "expected" is mostly "recent actual." Persistence is what makes the correlation look clean.
- Real rate stable ⇒ nominal rate moves **one-for-one** with inflation (Fisher)
  - **The Fisher effect holds (Pérez):** the inflation–nominal-rate correlation is ≈ **0.71** in US time series and ~**0.63** across countries. High-inflation countries have high nominal rates — Turkey/Argentina vs Switzerland.
  - **A caveat Abel stresses:** the real rate is **not constant** — negative in the mid-1970s, high in the early 1980s, negative again after 2008. So "i moves one-for-one with π" is a long-run tendency, not a short-run law.
- **Ex-ante vs ex-post.** The ex-ante real rate uses *expected* inflation: $r = i - \pi^e$. The ex-post one uses *realised*: $r = i - \pi$. They diverge when expectations miss. Example: lend at $i = 8\%$ expecting $\pi^e = 5\%$ (a planned 3% real return); if inflation turns out 9%, your **ex-post real return is −1%** — the borrower won. That gap is unexpected-inflation redistribution.

Figure 7.5 · Expected Inflation and Nominal Interest Rates (U.S.)



Source: Federal Reserve Bank of St. Louis (FRED).

The Fisher effect. The 10-year Treasury rate and survey expected inflation move together, $i = r + \pi^e$. Both came down from the early-1980s peak as inflation expectations were brought under control.

Take-away: the co-movement *confirms* Fisher, but the vertical gap — the real rate r — plainly *varies* (high in the early 1980s, negative after 2008), so "one-for-one" is a tendency, not a law.

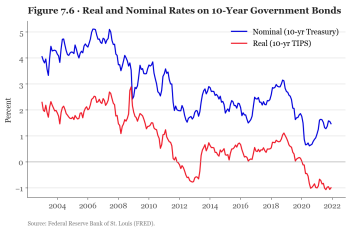
Data Application: the world real interest rate

The *ex-ante* real rate ($r = i - \pi^e$) is hard to measure because expected inflation is unobserved. Barro & Sala-i-Martin (1990) estimated a common real rate across the major economies and found it is *not* constant — it drifts with world saving and investment. That is the empirical counterpart to Abel's caveat: the Fisher line's intercept moves, so nominal rates do not track inflation one-for-one in the short run.

7.6.6 How can we measure inflation expectations?

- Find expectations ⇒ **surveys**, or **better**, read from **bond rates**
  - The survey route has names — Survey of Professional Forecasters, Livingston, Michigan — but the market route is usually more reliable, because there people back their beliefs with money.
- US issues **nominal bonds** and **TIPS**
- TIPS ⇒ **real** payments (interest & principal adjust for inflation)
  - **Treasury Inflation-Protected Securities (TIPS) don't fully protect you.** The US taxes the *inflation adjustment to principal* as income **in the year it accrues — before you receive the cash**. In high inflation the tax bill rises faster than the payout, so the bonds are less than perfect insurance.

- Compare the **nominal** rate with the **real** (TIPS) rate
  - The **gap** between the nominal and the TIPS yield is a *daily*, market-based read on **expected inflation** — no survey needed.
- **India:** the analogue is **IINSS-C** — inflation-indexed bonds with principal linked to **CPI**
  - **A cautionary tale:** launched 2013, the IINSS-C **largely flopped** — low uptake, thin trading — so India has no clean break-even read. The market-based measure needs a *liquid* indexed market, which not every country has.
- **History:** TIPS began in the US in **1997**, but indexed bonds are older — **Finland issued them in 1945**. Countries usually adopt them *after* an inflation scare.



Nominal vs. real (TIPS) yields. The 10-year nominal Treasury yield sits above the 10-year TIPS (real) yield; the vertical gap between them is the break-even inflation rate. Both fell after 2008, and the real yield turned negative.

Take-away: the market hands you two series — a real rate and a nominal rate — and their *difference* is expected inflation, with no survey required.

Data Application: surveys of inflation expectations

Before (and alongside) TIPS, expectations were read from surveys: the *Survey of Professional Forecasters* (since 1968), the *Livingston Survey* (since 1946), and the Michigan consumer survey. Surveys are smoother and slower-moving than the market's break-even; when the two disagree, it is usually because bond prices also carry risk and liquidity premia — so economists watch both.

7.6.7 The Break-Even Inflation Rate

Break-even = nominal-bond rate − real (TIPS) rate

- **Break-even** ⇒ rough **expected inflation**
  - Why "break-even": it is the inflation rate at which both bonds leave you equally well off. Above it, the nominal bond wins; below it, TIPS win.
- Inflation-risk premium ⇒ break-even **too high**; liquidity premium ⇒ **too low** ⇒ roughly **right**
  - **Two biases roughly cancel.** An **inflation-risk premium** makes investors pay extra for TIPS, pushing break-even **too high**; a **TIPS liquidity premium** (thinner market) pushes it **too low**. The net is small, so break-even is "roughly right," not exact.
- Data ⇒ break-even **fluctuates**; survey expectations fairly **constant**
- ⇒ markets differ from forecasters, **or** TIPS premia varied since 1998
  - The likely reading is *both* at once: markets are genuinely more forward-looking than forecasters *and* the two premia are themselves **time-varying**.
- **Tied to events:** break-even **collapsed in late 2008** and again in **March 2020**, when the market briefly priced in *deflation* — exactly when central banks launched QE. It doubles as a real-time fear gauge.



Break-even inflation. The gap between nominal and inflation-indexed (TIPS) yields is the market's expected inflation. It collapsed in late 2008, dipped again in 2020 — a real-time fear gauge — and rebounded in 2021.

Take-away: break-even swings far more than survey expectations — the *anomaly* is that it blends true expected inflation with time-varying risk and liquidity premia, so it is not a clean read of expectations.

Bottom line. Long-run inflation ≈ money growth − output growth, and expected inflation is built into the nominal rate through Fisher, $i = r + \pi^e$.

7.7 Taking Stock

7.7.1 Putting it all together

- **The chapter in four lines:** money demand $M^d = P\,L(Y,i)$; equilibrium $M/P = L$; quantity theory $\pi \approx g_M - g_Y$; Fisher $i = r + \pi^e$. Reproduce these four and you have Ch 7.
- **Money demand**: $L = P\,L(Y,i)$ — real demand **rises** with income, **falls** with the nominal rate
- **Asset-market equilibrium**: $\dfrac{M}{P} = L(Y,i) \;\Rightarrow\; P = \dfrac{M}{L(Y,i)}$
  - The hinge is still $M/P = L(Y,i)$ — it fixes the price level; the other three are it in growth rates or extended to expectations.
- **Quantity theory** ($MV = PY$): in the long run, **inflation ≈ money growth − real output growth**
- **Fisher relation**: $i = r + \pi^e$ — expected inflation is built into nominal rates
- **The deepest idea — money neutrality / the classical dichotomy:** in the long run, real variables (output, real wage, r) are set by technology and resources, while nominal variables (P, π, i) are set by money. The two do not interact.
- **A myth, corrected with data (Pérez):** "inflation makes us poorer because wages don't keep up." Over 1965–2015, US **nominal wages rose almost one-for-one with the CPI**; real wages followed productivity, not inflation. Real wage = MRPL (marginal revenue product of labour). Inflation's true costs are subtler — shoeleather, menu, distortions, redistribution — not a general fall in living standards.
- **A short-run caveat:** wages and prices *are* sticky in the short run, so moderate inflation does redistribute. The layman's view is not wrong month-to-month, only in the long run.

7.7.2 Building blocks: how the pieces connect

Portfolio choice → Money demand $L(Y,i)$ → Asset-market equilibrium $\dfrac{M}{P} = L$ → Price level → Money growth ⇒ inflation → Nominal rates ($i = r + \pi^e$)

- **The flow is the chapter order:** portfolio choice → money demand $L(Y,i)$ → equilibrium $M/P = L$ → price level → money growth ⇒ inflation → nominal rates via Fisher. Note the **feedback loop** (Pérez): the nominal rate loops *back* into money demand, closing the system.
- The asset market is the **third leg** of the Ch 9 three-market model (labor · goods · **assets**)

  - Its job in that model: it supplies the **nominal side** (P, π), while labour and goods supply the **real side** ($\bar{Y}$, r).
- It pins down the **nominal** side — the **price level and inflation** — while the goods and labor markets set the **real** side
- **What's next:** in Ch 9 we stop passing r and P between markets one at a time and solve all three **together** (**IS–LM–FE**: goods, money, and full-employment [FE] labour markets). This chapter built the asset leg; next chapter wires it to the rest.

Economists behind this chapter

Jean Bodin (1530–1596, French) — tied Europe's 16th-century "price revolution" to the inflow of New World silver; an early statement of the quantity theory of money.

David Hume (1711–1776, Scottish) — essay *Of Money* (1752); the quantity theory and the long-run *neutrality* of money (double the money, double all prices).

Knut Wicksell (1851–1926, Swedish) — the natural vs. market rate of interest and the cumulative process linking money, credit, and inflation.

Irving Fisher (1867–1947, American) — *The Theory of Interest* (1930); the Fisher relation $i = r + \pi^e$ and the equation of exchange $MV = PT$.

John Maynard Keynes (1883–1946, British) — *The General Theory* (1936); the liquidity-preference theory of money demand (transactions, precautionary, speculative motives).

B. R. Ambedkar (1891–1956, Indian) — *The Problem of the Rupee* (1923); monetary economics of the rupee and the exchange standard, arguing for *price stability* over a fixed exchange rate; his analysis fed into the conception of the Reserve Bank of India.

Milton Friedman (1912–2006, American) — restated the quantity theory and monetarism; "inflation is always and everywhere a monetary phenomenon"; Nobel 1976.

James Tobin (1918–2002, American) — portfolio theory of money demand and the Baumol–Tobin model; risk and asset choice; Nobel 1981.

William Baumol (1922–2017, American) — the inventory (Baumol–Tobin) theory of the transactions demand for money (1952).

Harry Markowitz (1927–2023, American) — modern portfolio theory: the risk–return trade-off and diversification ("Portfolio Selection," 1952); Nobel 1990.

Phillip Cagan (1927–2012, American) — the monetary dynamics of hyperinflation (1956) and the seigniorage "Laffer curve."

C. Rangarajan (b. 1932, Indian) — Reserve Bank of India Governor; built India's "monetary targeting with feedback" and its money-supply analytics.

Sukhamoy Chakravarty (1934–1990, Indian) — chaired the 1985 committee that recommended *monetary targeting* (M3 growth) for India.

Raghuram Rajan (b. 1963, Indian) — RBI Governor; architect of India's flexible *inflation-targeting* framework (the CPI 4% ±2 band) and the Monetary Policy Committee.

Modern Monetary Theory (heterodox school; W. Mosler, L. R. Wray, S. Kelton) — holds that a government issuing its own fiat currency faces an *inflation* constraint, not a solvency one, so deficits are bounded by real resources; a contemporary challenge to the quantity-theory and "monetising the deficit" framing used in this chapter.

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***** END *****

In theory, theory and practice are the same. In practice, they are not.

— Anonymous

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